

Neon lamp

Y13 (2013) — English translation

The electrical circuit (Fig. 1) consists of an ideal inductor L capacitor C , a battery with EMF U_0 and a nonlinear NE element, the current-voltage characteristic of which is shown in Fig. 2 (gas-filled lamps have a similar characteristic). It is known that the hysteresis zone $\Delta U \ll U$; in addition current $I_0 \gg U_0 \sqrt{\frac{C}{L}}$

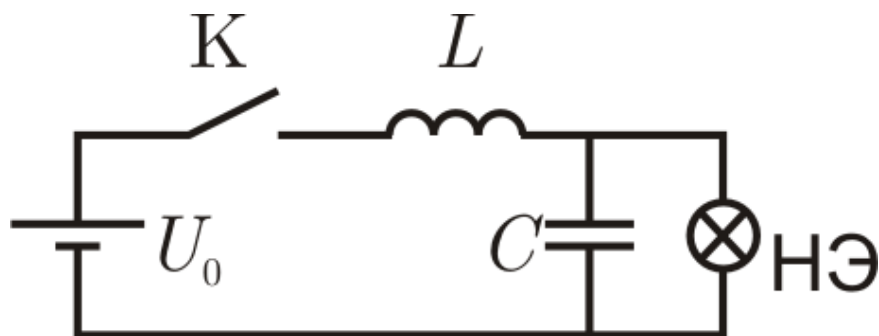


Figure 1: Figure 1.

Rice. 2. Current-voltage characteristic of a nonlinear element

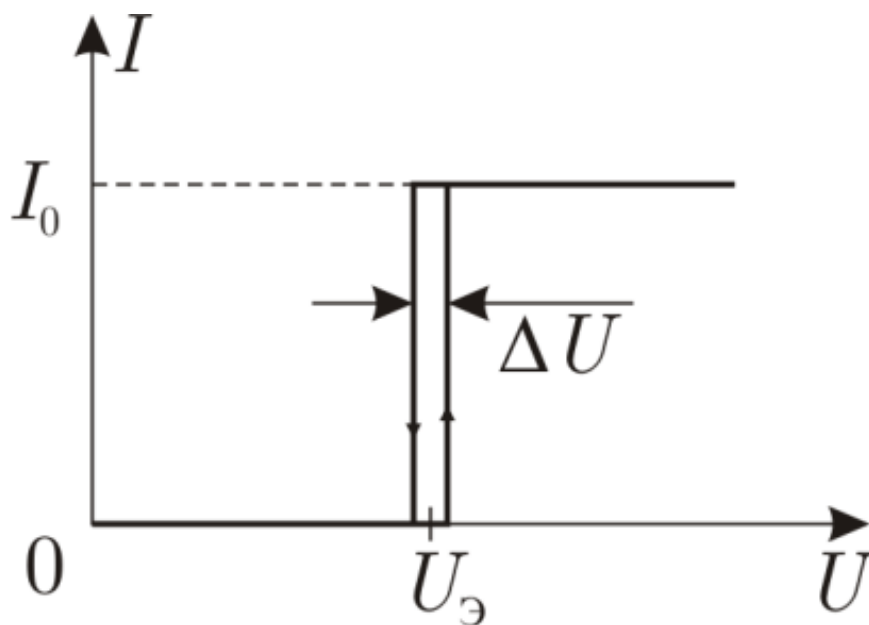


Figure 2: Figure 2.

A1 Determine how many times the lamp will flash after closing the key K , if $U_0 < U < 2U_0$. Draw (qualitatively) the process on the graph $U_C(t)$.

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A2	Determine the steady-state flash frequency f and the average power W dissipated in the lamp if $U_0 = U$	0
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Source: <https://pho.rs/p/3975>